

Ly Vireak Dara, MSc

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PROFESSIONAL SUMMARY

Computer vision researcher specializing in open-vocabulary object detection and multi-object tracking robustness. First-author on YOLO-World architecture improvements (EnYOLO-World, +3.1 AP) and lead researcher on VocabDriftMOT — the first benchmark for vocabulary-drift failure modes in OV-MOT (510 annotated events, 21 sequences). 4 years production ML engineering experience.

EDUCATION

M.S. Artificial Intelligence | Xidian University, Xi'an, China Sep 2023 – Sep 2026 (Expected)

Thesis: Enhanced Open-Vocabulary Object Detection using YOLO Architectures

B.A. Computer Science and Engineering | Royal University of Phnom Penh, Cambodia Sep 2015 – Aug 2019

SKILLS

Programming: Python, JavaScript, Java, TypeScript, SQL

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, YOLO Series (v5-v11), CLIP, ByteTrack

Computer Vision: OpenCV, TorchVision, Object Detection, Image Segmentation, ONNX

MLOps & Cloud: Docker, MLflow, FastAPI, Redis, GCP Cloud Run, Prometheus, Grafana, Git, REST APIs

Data & RAG: Pandas, NumPy, PostgreSQL, MongoDB, LangChain, ChromaDB, FAISS, Cross-Encoder Reranking

WORK EXPERIENCE

Software Engineer | Korea Software Innovation Global Network (KOSIGN) Oct 2020 – Aug 2023

- Developed and maintained full-stack applications serving 10,000+ users across Korea and Southeast Asia
- Designed PostgreSQL database schemas and optimized queries, reducing API response times by 40%
- Built RESTful APIs with Spring Boot handling 500+ requests/second with 99.9% uptime
- Collaborated with Korean development team using Agile methodology, delivering 12 product releases

Software Engineer | Web Essentials Oct 2019 – Aug 2020

- Built responsive front-end interfaces using React.js, improving user engagement by 25%
- Integrated third-party APIs for payment processing and authentication systems
- Implemented comprehensive testing procedures reducing production bugs by 30%

AI/ML PROJECTS

Multimodal Visual RAG System 2026

- End-to-end image search pipeline: CLIP embeddings + ChromaDB vector store + cross-encoder reranker + LLaMA 3 answer generation
- Qwen2.5-VL-72B auto-describes uploaded images using vision AI — no manual tagging required
- REST API (FastAPI) with /index, /query, /reset endpoints; Docker containerized, deployed live on GCP Cloud Run
- GitHub: github.com/Vireakdara/multimodal-rag | Live: multimodal-rag-322269723789.asia-southeast1.run.app

VocabDriftMOT: Vocabulary-Drift Benchmark for OV-MOT 2026

- First benchmark for mid-stream vocabulary switching in open-vocabulary multi-object tracking; 510 annotated transition events across 21 MOT17 sequences and 3 detector variants
- Identified detector-agnostic failure: synonym transitions cause near-complete identity loss (IDF1-AT = 0.009) vs hypernym expansion (0.369) — 40x gap confirmed across all detector variants
- Introduced IDF1-AT and SJS metrics; designed PromptBridge V2 geometric gate (70% correct preserve/terminate decisions)
- GitHub: github.com/Vireakdara/VocabDriftMOT

OpenDet-Bench: Production Open-Vocabulary Detection Platform 2026

- End-to-end detection and tracking pipeline: YOLOv8 + ByteTrack + ONNX (1.54x inference speedup) + MLflow + Docker + GCP Cloud Run
- Real-time video analytics with FastAPI backend, Redis buffer, WebSocket dashboard, Prometheus/Grafana monitoring

- Live public API deployed on GCP Cloud Run

EnYOLO-World: Enhanced Open-Vocabulary Object Detection

2025

- Developed novel PGI (Programmable Gradient Information) integration with YOLO-World architecture
- Designed FiLM-PAN module for dynamic feature modulation, achieving +3.1 AP over baseline
- Evaluated on 5+ benchmark datasets (LVIS, COCO, Objects365) using PyTorch

Vision-Language Models (CLIP) & Prompt Engineering

2024

- Fine-tuned CLIP via Hugging Face for zero-shot classification on domain-specific datasets
- Engineered context-aware prompts for medical imagery/text alignment tasks
- Integrated FAISS vector database for semantic search, accelerating retrieval by 60%

PUBLICATIONS

EnYOLO-World: Enhanced Open-Vocabulary Detection with PGI and FiLM-PAN

ELCVIA (Under Review) · First Author

Fine-grained Meta-learning with Semantic Augmentation for SAR Change Detection

Journal of Applied Remote Sensing, Oct 2025 · Co-Author

Non-Rigid Registration Based on Rotation-Invariant Feature Flow

IEEE IGARSS 2025 · Co-Author

An Edge-Enhanced Siamese Network for 3D Change Detection

IEEE IGARSS 2025 · Co-Author

LANGUAGES

Khmer (Native) · English (Fluent) · Chinese (HSK3)